

Submission Summary

Conference Name

2nd International Conference on Computational Intelligence and Information Retrieval

Paper ID

490

Paper Title

Drug Inventory and Supply Chain Tracking System using Blockchain

Abstract

The system records all drug-related transactions such as manufacturing, distribution, and dispensing in a secure, append-only ledger, where each entry is cryptographically linked to its predecessor to ensure immutability. The inventory details are kept in a detailed stock record in an SQL Server/MySQL database (off-chain storage), and the SHA-256 digests are stored in the ledger- style blockchain to provide the tamper-evidence and traceability. We implement the business logic and APIs in Java so that the application is portable and scalable. The hybrid architecture design helps to increase the trust, enhances the transparency between different stakeholders, and enables the instant drug authenticity verification, which can reduce the counterfeit drugs, ensure compliance with regulatory requirements.

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Primary Subject Area

Advance Computing

Submission Files

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